



Hospital Management System – SQL Database Project

Total points: 100 + 20 bonus

Project Objective

This project simulates a **Hospital Management System** using a relational SQL database. In the first part of this project, you will **set up the hospital management database** by defining the structure of each table and the data relationships between tables. In the second part, you will **write queries on this database** to answer questions and gain insights. The data for each table is provided to you in your **project_data.sql** file.

Tables in the Dataset

1. hospitals

- Contains general information about hospitals in the system.
 - **Key Fields:** `hospital_id`, `name`, `address`, `phone`
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2. doctors

- Stores details about doctors and their hospital affiliations.
 - **Key Fields:** `doctor_id`, `name`, `specialty`, `hospital_id`, `email`
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3. patients

- Holds patient demographics and contact information.
 - **Key Fields:** `patient_id`, `name`, `dob`, `address`, `phone`
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4. appointments

- Tracks appointments between patients and doctors.
 - **Key Fields:** `appointment_id`, `patient_id`, `doctor_id`, `appointment_date`, `reason`
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5. departments

- Represents medical departments within hospitals.
 - **Key Fields:** `department_id`, `hospital_id`, `name`
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6. medications

- Contains the list of available or prescribed medications.
 - **Key Fields:** `medication_id`, `name`, `description`
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7. prescriptions

- Links doctors, patients, and medications together.
 - **Key Fields:** `prescription_id`, `patient_id`, `doctor_id`, `medication_id`, `prescribed_date`
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8. rooms

- Stores information about rooms assigned to departments.
 - **Key Fields:** `room_id`, `room_no`, `department_id`, `capacity`
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Part 1: Database Setup and Data Insertion (30 points)

Set up the hospital management database by defining the structure of each table and inserting data into it from your **project_data** file. A critical part of this phase is **establishing data relationships** through primary keys (PK) and foreign keys (FK), ensuring referential integrity between tables for e.g. linking doctors to hospitals, appointments to patients and doctors, and prescriptions to medications. This prepares the database for advanced querying and data analysis.

Key tasks:

- Create a new schema for your project and create all tables inside it.
- Write **CREATE TABLE** statements for all the tables keeping the following in mind:
 - Define appropriate data types for each column (e.g. int, text, date).
 - Specify primary keys to uniquely identify records.
 - Add foreign keys to represent relationships between tables.

- Insert data into each table using **INSERT** statements from the **project_data** file.
 - Validate referential integrity by ensuring foreign keys correctly reference primary keys.
 - Write simple **SELECT** queries to confirm data insertion.
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Part 2: SQL Queries to Analyze Hospital Data (70 points)

Write SQL queries to answer the following questions based on the database. These queries will include the use of all covered concepts like filtering, sorting, aggregations, joins, subqueries, Common Table Expressions (CTEs), window functions, and conditional logic using CASE statements.

Queries Part 1 (3 points each):

- Find all patients born after the year 2000
- Find all prescriptions for patient_id 5
- Show the number of appointments per month
- List all medications that include the keywords pain or infection in the description
- Show all doctors with their hospital name
- Find the names, phones and appointment_dates of all patients with appointments in August 2025
- Show all room numbers and their capacities from the Neurology departments
- Count how many doctors work in each hospital
- Find patients who have more than 3 appointments
- List appointments with patient and doctor names

Queries Part 2 (5 points each)

- For all the **emergency** appointments, show patient name, date of birth as well as age-group:
 - 18 or below - Pediatric
 - 19 to 64 - Adult

- 65 or above - Geriatric
 - Show all departments of Green Valley Medical Center
 - Find patients who have never had a prescription
 - List name, address and phone for patients who have appointments in more than one hospital
 - Show the latest appointment for each patient
 - Show the 3rd most frequently prescribed medication/medications
 - Show all hospitals with the lowest doctor count
 - Find the department with the second largest room capacity in each hospital
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Project Deliverables

- A report containing:
 - SQL scripts for each task. **(100 points)**
 - Any further insights derived from the data for example: **(Up to 20 bonus points)**
 - Top hospitals with the most specialists in a particular field (e.g., cardiology)
 - Average room capacity per department or per hospital.
 - The trend of appointments over time (daily/weekly/monthly).
 - Most frequent appointment reasons.
 - Data visualizations using tools like Tableau or Power BI
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